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Are all eigenvectors, of any matrix, always orthogonal?

Asked 10 years, 4 months ago Modified 2 months ago Viewed 81k times

68

32

I have a very simple question that can be stated without proof. Are all eigenvectors, of any matrix, always orthogonal? I am trying to understand Principal components and it is cruucial for me to see the basis of eigenvectors.

linear-algebra matrices eigenvalues-eigenvectors

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edited Nov 26, 2019 at 14:30
zabop
955 6 18

asked May 8, 2012 at 13:30
Bober02
2,368 3 16 15

9 No. Take any any non-orthogonal basis $(v_1,\ldots,v_n)$ and define a linear map A on this basis by sending each v_i to iv_i. The eigenspaces are the n lines generated by the v_i, and these are by construction not ortgogonal. – Olivier Bégassat May 8, 2012 at 13:38

13 Eigenvectors corresponding to different eigenvalues will be orthogonal if the matrix is symmetric. This is part of the real spectral theorem. – Dylan Moreland May 8, 2012 at 13:38

2 The case @Dylan is describing will apply to your study of principal components, since the underlying matrices are symmetric... – J. M. ain't a mathematician May 8, 2012 at 14:50

@PL this is clearly not a duplicate of that question - this question is about the general case when M is not necessarily a real symmetric matrix. – KReiser Jul 26, 2020 at 1:55

4 Answers

Sorted by: Highest score (default)

50

In general, for any matrix, the eigenvectors are NOT always orthogonal. But for a special type of matrix, symmetric matrix, the eigenvalues are always real and the corresponding eigenvectors are always orthogonal.

For any matrix M with n rows and m columns, M multiplies with its transpose, either M*M' or M'M, results in a symmetric matrix, so for this symmetric matrix, the eigenvectors are always orthogonal.

In the application of PCA, a dataset of n samples with m features is usually represented in a n* m matrix D. The variance and covariance among those m features can be represented by a m*m matrix D'*D, which is symmetric (numbers on the diagonal represent the variance of each single feature, and the number on row i column j represents the covariance between

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answered Feb 21, 2017 at 6:15

Yue Tyler Jin
601 5 2

28

Fix two linearly independent vectors u and v in $\mathbb{R}^2$, define $Tu = u$ and $Tv = 2v$. Then extend linearly T to a map from $\mathbb{R}^n$ to itself. The eigenvectors of T are u and v (or any multiple). Of course, u need not be perpendicular to v.

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answered May 8, 2012 at 13:40

Siminore
34k 3 49 78

Thank you very much for your answers. Could anyone state whether they are orthogonal in PCA case? – Bober02 May 8, 2012 at 13:45

2 For PCA, things can always be set up such that the eigenvectors are orthogonal. On the other hand, I would recommend looking at PCA as a singular value decomposition instead of as an eigendecomposition. It's been discussed here on math.SE a number of times; search around. – J. M. ain't a mathematician May 8, 2012 at 14:48

What do you mean by "setting up"? Is there some common technique to achive a singular matrix? – Bober02 May 9, 2012 at 8:12

3 My understanding based on en.wikipedia.org/wiki/Principal_component_analysis is that a singular value decomposition (SVD) of X results in three matrices, where the first of them is composed of the eigenvectors of $X^T X$, and is used for PCA. This matrix is always orthogonal. – Uri Jun 14, 2013 at 13:04

Beautiful answer. – Don Larynx Nov 13, 2013 at 15:14

4

In the context of PCA: it is usually applied to a positive semi-definite matrix, such as a matrix cross product, $X'X$, or a covariance or correlation matrix.

In this PSD case, all eigenvalues, $\lambda_i \geq 0$ and if $\lambda_i \neq \lambda_j$, then the corresponding eivenvectors are orthogonal. If $\lambda_i = \lambda_j$ then any two orthogonal vectors serve as eigenvectors for that subspace.

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answered Jan 9, 2017 at 4:00

user101089
449 5 10

Not necessarily all orthogonal. However two eigenvectors corresponding to different eigenvalues are orthogonal, whenever the matrix is symmetric.

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Now,

$AX_1 = \lambda_1 X_1$ and $AX_2 = \lambda_2 X_2$.

Taking Transpose of first,

$$\begin{aligned} & (AX_1)^T = (\lambda_1 X_1)^T \\ \implies & X_1^T A^T = \lambda_1 X_1^T \\ \implies & X_1^T A^T X_2 = \lambda_1 X_1^T X_2 \\ \implies & X_1^T \lambda_2 X_2 = \lambda_1 X_1^T X_2 \\ \implies & \lambda_2 X_1^T X_2 = \lambda_1 X_1^T X_2 \\ \implies & (\lambda_2 - \lambda_1) X_1^T X_2 = 0 \end{aligned}$$

Since $\lambda_2 \neq \lambda_1$, $X_1^T X_2$ must be 0. This establishes orthogonality of eigenvectors corresponding to two different eigenvalues.

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edited Jul 29 at 13:58
Marine Galantin
2,785 1 8 29

answered Jul 17, 2021 at 4:53
Jai Vrat Singh
45 3

5 It's implicitly assumed that A is symmetric. – FuzzyDuck Dec 24, 2021 at 18:32